



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Term Final Exam, Fall 2025
CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all of the questions. Figures in the right-hand margin indicates full marks.
Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Read the following code and detect different types of errors.

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1 #include <stdio.h> 2 #inclade <stdlib.h> 3 #include <String.h> 4 void faultyFunction(int n) { 5 int arr[3]; 6 arr[5] = n; 7 return "done"; 8 }	15 int y = 5 / 0; 16 char buf[5]; 17 strcpy(buff, "TooLong"); 18 int *p = NULL; 19 *p = 42; 20 *p = a>buff[0] ? 10 : a; 21 faultyFunction(p); 22 int mem = malloc(10 * sizeof(int)); 23 mem[0] = 1; 24 return 0 }
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2. Generate Recursive Predictive Parsing Table for the following grammar with FIRST and FOLLOW sets.

Check using parsing table whether string “1+1*23+(3+21*(2))” is accepted or not.

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$S \rightarrow E$

$E \rightarrow E + T \mid T \mid \epsilon$

$T \rightarrow T * F \mid F \mid \epsilon$

$F \rightarrow (E) \mid FF \mid 1 \mid 2 \mid 3$

3. Eliminate immediate left recursion from the following grammar.

3*2

(a) $S \rightarrow S+S \mid (S) \mid SSA \mid *AS$ $A \rightarrow A \mid AAc \mid cca \mid Aab$ $B \rightarrow BA+ \mid BaB \mid B(BA) \mid b \mid \epsilon$ $D \rightarrow DC \mid a/C \mid DDD/C \mid \epsilon$ $C \rightarrow (C) \mid a \mid b \mid cC \mid CS$	(b) $S \rightarrow S+A \mid SB*C \mid T/SD \mid \epsilon$ $A \rightarrow A-2B \mid C*A \mid BA \mid \epsilon$ $B \rightarrow B^{\wedge}C \mid D*B \mid A/B \mid b$ $C \rightarrow ADE \mid EF*C \mid F$ $D \rightarrow D+S \mid d \mid \epsilon$ $E \rightarrow E*E \mid A+D \mid e$ $F \rightarrow (S) \mid f$
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4. Left Factor the following. [(a) consider each character & (b) consider as word]

3*2

- a. $A \rightarrow (ab+b(1a)) \mid (aa+b(1)) \mid (ab-(a1)) \mid ((1)) \mid (aa+b(2)) \mid (ab-(1)+1) \mid (ab + 1(a)) \mid (ab+b(1b)) \mid ((a+b)+1) \mid (a+21) \mid (ab-1+2) \mid \epsilon$
- b. $A \rightarrow \text{Sun Rises} \mid \text{Sun Rising} \mid \text{San Rays} \mid \text{Sun Raying} \mid \text{Sun Resting} \mid \text{Sun Reflection} \mid \text{Sun Reflected} \mid \text{Moon Rises} \mid \text{Moon Rising} \mid \text{Moon Rays} \mid \text{Moon Raying} \mid \text{Moons Resting} \mid \text{Moon Reflection} \mid \text{Moons Reflected} \mid \text{Moonlight} \mid \text{Moonlit}$

5. For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. Consider all the terminals are of single character/number/symbol.

4*3

(a) $A \rightarrow CBA \mid d \mid eBfA$ $B \rightarrow AC+ \mid gh \mid BA^*t \mid \epsilon$ $C \rightarrow d \mid FAE$ $E \rightarrow pop \mid F \mid \epsilon$ $F \rightarrow keyC \mid hEAF \mid \epsilon \mid y$	(b) $S \rightarrow TUS \mid n \mid nUoS$ $U \rightarrow ST/ \mid pq \mid US+Vr$ $T \rightarrow m \mid VUR$ $R \rightarrow cat \mid RV$ $V \rightarrow dogT \mid qRUZ \mid z \mid \epsilon$
(c) $A \rightarrow mABU+s \mid jAVrq \mid BUsrq \mid Vmn \mid ef$ $B \rightarrow rsA^*V \mid e \mid f \mid Vjk \mid UopAg \mid \epsilon$ $V \rightarrow Bp \mid VdUBq \mid jpAqfBAZ/ \mid eB \mid \epsilon$ $U \rightarrow m \mid n \mid BV+ \mid jBrUV \mid \epsilon$	